Rishi More

Boston, MA | rishimore102@gmail.com | +1 (650) 220-9450 | linkedin.com/in/rishimore102 | github.com/rishi-more-2003 | rishi-more-2003.github.io

Summary

ML systems engineer specializing in efficient and reliable LLM inference. **ICML 2026 co-first author** with hands-on experience in CUDA/Triton kernels, Nsight profiling, INT4 quantization, vLLM, post-training evaluation, and distribution-free risk control.

Experience

JHU Data Science and AI Institute (DSAI) — **May 2025 – June 2026**
Graduate Research Assistant, LLM Inference Optimization — *Baltimore, Maryland*

- Co-first authored *Conformal Thinking* (**ICML 2026 main track**), introducing finite-sample calibrated early stopping for reasoning LLMs while controlling a target error rate.
- Built a vLLM-based calibration and evaluation pipeline across **4 reasoning models**, **4 benchmarks**, and **2.7K+ problems**, maintaining target error rates under dataset and response-length shift with finite-sample UCB calibration.
- Designed dual-threshold and ensembled halt policies that reduced reasoning tokens by **12% at matched accuracy**, with reusable tooling for calibration, evaluation, and ablation studies.

JHU Center for Language and Speech Processing (CLSP) — **January 2025 – December 2025**
Graduate Researcher, Retrieval Systems — *Baltimore, Maryland*

- Built temporal RAG over **10K+ timestamped documents** using revision-aware chunk tracking, time-decay scoring, and metadata-scoped indexing to prioritize current evidence.
- Improved retrieval precision by **23%** over dense-only retrieval and reduced stale or hallucination-prone contexts by **31%** through recency weighting and reference-free consistency scoring.

Mastek — **November 2022 – July 2023**
Machine Learning Engineer Intern — *Mumbai, India*

- Deployed an on-premises edge voice-authentication system with **<100 ms p99 latency** and **96.6% accuracy**, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment.
- Built a TensorFlow pipeline over **1M+ voice samples**; spectral-gating denoising, augmentation, and robust feature extraction improved authentication reliability by **20%** in high-noise conditions.

Projects

Conformal Serve: Risk-Controlled Adaptive LLM Serving | *vLLM, FastAPI, Docker Compose, Prometheus*

- Built an OpenAI-compatible inference server over vLLM's AsyncLLMEngine, preserving KV-cache reuse while early-stopping reasoning at client-specified risk levels α.
- On Qwen2.5-1.5B-Instruct and an 8 GB RTX 4070, reduced reasoning tokens **64.7% at** $\alpha = 0.10$ and **71.4% at** $\alpha = 0.30$ while maintaining realized error $\leq \alpha$ on small held-out arithmetic sets.
- Calibrated per-α thresholds offline by distribution-free risk control into immutable, content-addressed artifacts; shipped Docker Compose with Prometheus/Grafana and a one-command benchmark reporting **95%** Clopper–Pearson error bounds.

INT4 KV-Cache Quantization & Fused Triton Attention for Long-Context Decode | *Triton, CUDA, PyTorch*

- Built a correctness-gated fused Triton INT4 dequantization-attention kernel, cutting 32K-context attention latency **13.7**× (**7.54 to 0.55 ms/layer**) versus PyTorch FP16 GQA SDPA while maintaining **0.999999 cosine similarity** to the dequantized oracle.
- Reduced KV storage from **28.7 KB to 8.3 KB per token** (**3.47**×), raising estimated 8GB KV capacity from ∼**154K to** ∼**580K tokens**; retained **100% passkey retrieval through 32K** with only **+1.0% WikiText perplexity**.
- Diagnosed per-token-K failure as outlier-channel collapse (attention cosine **0.08 vs. 0.98**) and built an RCPS controller certifying ∼**50% cache eviction** at **19% realized loss** under a **30% risk budget**.

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | *CUDA C++, Nsight, vLLM*

- Profiled Qwen2.5-1.5B batch-1 decode with Nsight; attributed ∼**81% of GPU-kernel time** to weight GEMVs and measured dominant MLP projections at ∼**247 GB/s** against a ∼**250 GB/s** DRAM ceiling.
- Built a correctness-gated fused CUDA INT4 group-dequantization + GEMV kernel reaching ∼**223 GB/s**; accelerated batch-1 MLP projections by **2.38**× versus FP16 GEMM and mapped the crossover to batch 2 as weight reuse made GEMM faster.

Education

Johns Hopkins University — **May 2026**
M.S.E. in Computer Science — *GPA: 3.9/4.0*

University of Mumbai — **July 2024**
B.Tech. in Computer Science, Honors in AI/ML — *CGPA: 9.85/10*

Technical Skills

Languages: Python, C++, CUDA C++, C, SQL, Bash
ML/LLM Systems: PyTorch, Triton, vLLM, Hugging Face Transformers, TensorFlow, LoRA, DPO, RAG
Performance & Infrastructure: Nsight Compute/Systems, Linux, Git, Docker, GitHub Actions, AWS
Methods: Quantization, kernel fusion, roofline analysis, LLM evaluation, conformal risk control, calibration